

Parametric Heatsink Conjugate Heat Transfer Simulation Report

OpenFOAM 13 · foamMultiRun (Buoyant, Multi-Region) · Parallel/Serial Verified
20,000 Steady-State Iterations

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Parameter	Value
Solver	foamMultiRun (buoyant, 3-region CHT)
Regions	fluid (air); solid → heatsink (Al) + cpu (Si) cell zones
Total mesh cells	649,200 (577,200 fluid + 72,000 solid)
Total iterations	20,000 (steady-state)
CPU heat dissipation	150 W
Fan	Delta FFB0812VH (quadratic curve, fanPressure BC)
Fan outlet velocity (converged)	3.87 m/s
Fin pitch / fin count (baseline)	5 mm / 15 fins
Max CPU temperature	343.90 K (70.9°C)
Inlet air temperature	300 K (27°C)
Verification	Serial run vs. 8-core parallel run agree to 0.001 K

1. Project Overview

This project presents a steady-state Conjugate Heat Transfer (CHT) simulation of a CPU and fin-array heatsink inside a fan-driven duct, built with OpenFOAM 13's foamMultiRun multi-region driver. Unlike a single fixed-geometry case, the mesh itself is generated parametrically: a Python script lays out the fin/channel geometry directly as a blockMeshDict from a fin pitch and fin count, so the design can be swept without rebuilding CAD geometry by hand. The goal is to predict CPU junction temperature under 150 W dissipation with fan-curve-driven (rather than fixed-velocity) airflow, and to use that parametric capability to find the thermal optimum fin pitch for a given fan/duct combination.

Key Features

- 3-region CHT: fluid air domain, aluminium heatsink, and silicon CPU die, coupled via coupledTemperature interfaces
- Fully parametric blockMeshDict generation (gen_cases_v2.py) — fin pitch and fin count are input parameters, not hand-edited geometry
- Fan modelled with a fanPressure boundary condition driven by a tabulated quadratic fan curve (Delta FFB0812VH), not a fixed-velocity inlet
- Volumetric heatSource fvModel applies 150 W directly to the CPU silicon cell zone
- Baseline case cross-checked on 1 core (serial) and 8 MPI ranks — converged results agree to 0.001 K
- 20,000-iteration steady-state run monitored via cpuTmax / solidTmax volume and surface field-value function objects

2. Geometry and Mesh

2.1 Parametric Geometry Generation

The heatsink/duct geometry is produced directly as an OpenFOAM blockMeshDict by gen_cases_v2.py, rather than imported from CAD. Given a fin pitch and fin count, the script lays out alternating fin / channel / side-gap strips across the 80 × 80 mm duct cross-section, splits those strips against the CPU footprint, and writes the resulting hex-block mesh with fan, wall, and region-zone tagging already built in. This makes a pitch sweep a one-line parameter change rather than a CAD rebuild — the central point of the study.

Fixed geometry: duct cross-section 80 × 80 mm; fin thickness 1.5 mm; fin height 35 mm; base thickness 5 mm; CPU die 40 × 40 × 4.5 mm centred under the fin array; inlet and outlet plenum length 105 mm each (total duct length 290 mm). The baseline case presented here uses a 5 mm fin pitch, giving a 3.5 mm channel width, 15 fins, and 4.25 mm side gaps.

2.2 Mesh

The block-structured hex mesh is built with blockMesh and split into coupled regions using splitMeshRegions -cellZones, producing 649,200 cells in total across the fluid and solid domains.

Region	Material	Cells	Role
fluid	Air (buoyant, k-ε)	577,200	Flow & heat transport
heatsink	Aluminium (k = 205 W/m·K, ρ = 2700)	~70,800	Heat spreading / dissipation
cpu	Silicon (k = 150 W/m·K, ρ = 2329)	1,200	Heat source (150 W)

3. Simulation Setup

3.1 Solver Configuration

The case is driven by foamMultiRun, coupling one buoyant fluid region (solved with k-ε turbulence and a p_rgh pressure formulation) to a solid super-region containing the heatsink and cpu cell zones. Region interfaces use the coupledTemperature boundary condition at every fluid↔heatsink, fluid↔cpu, and heatsink↔cpu face. Heat is injected directly into the CPU silicon zone using a heatSource fvModel (cellZone allCpuCells, Q = 150 W) rather than a converted volumetric flux boundary condition.

3.2 Fan Boundary Condition

Rather than a fixed-velocity inlet, the fan is modelled with a fanPressure boundary condition at the fan1_half0 patch, driven by a tabulated pressureVsQ.csv curve. The table is generated from the quadratic Delta FFB0812VH fan curve $dP = 73.99 - 1164.25 \cdot Q - 34615 \cdot Q^2$ [Pa, Q in m³/s], so the operating point of the fan against the duct's actual flow resistance is solved for, not assumed.

3.3 Boundary Conditions Summary

Boundary	Field	Type	Value
fan1_half0 (inlet)	p_rgh	fanPressure	Delta FFB0812VH curve (table)
fan1_half0 / fan2_half0	U	pressureInletOutletVelocity	Solved from fan curve
base_bottom / top_wall / side_walls	U	noSlip	—
fluid↔heatsink, fluid↔cpu, heatsink↔cpu		coupledTemperature	Region-coupled
CPU cell zone	heatSource (fvModel)	heatSource	150 W

3.4 Verification: Serial vs. Parallel

The baseline 5 mm-pitch case was run twice to 20,000 iterations — once on a single core and once decomposed across 8 MPI ranks — to confirm that domain decomposition does not change the converged solution. Both runs converge to $T_{\text{max}} = 343.900$ K, agreeing to within 0.001 K, which is a standard cross-check before trusting a decomposed run for a larger parametric sweep.

4. Results

4.1 Convergence History

CPU maximum temperature, monitored via the `cpuTmax volFieldValue` function object, settles to 343.90 K by approximately 15,000 iterations in both the serial and parallel runs. By iteration 20,000, residuals for `U`, `h`, `e` (energy in both solid regions), and `p_rgh` are all below 1×10^{-10} , indicating a well-converged steady state.

4.2 Thermal Performance

At 150 W CPU dissipation and the 5 mm/15-fin baseline geometry, the duct reaches a converged fan exit velocity of 3.87 m/s. The CPU die settles at 343.90 K (70.9°C) against a 300 K (27°C) inlet — a CPU-to-ambient temperature rise of approximately 44 K under fan-curve-driven airflow, rather than an assumed fixed flow rate.

4.3 Workflow

`gen_cases_v2.py` (parametric `blockMeshDict` generator) → `blockMesh` (hex mesh with fin/channel/side-gap strips and fan/wall patches) → `splitMeshRegions -cellZones` (fluid | heatsink | cpu) → `foamMultiRun` (coupled buoyant + conductive solve, 20,000 iterations) → `postProcessing` (`cpuTmax`, `solidTmax`, fan patch flow rate and velocity).

5. Status and Future Work

This baseline case is the first point of an intended three-point fin-pitch sweep (4 mm / 5 mm / 6 mm), aimed at finding the thermal optimum for this fan and duct combination: tighter fin pitch increases flow resistance and reduces airflow, while wider pitch reduces total fin surface area — both pushing CPU temperature up, with a minimum expected at an intermediate pitch.

Case	Status	Result
5 mm pitch, 15 fins (serial)	Converged, 20,000 it.	$T_{\text{max}} = 343.90 \text{ K}$
5 mm pitch, 15 fins (8-core parallel)	Converged, 20,000 it.	$T_{\text{max}} = 343.90 \text{ K}$
4 mm pitch (18 fins)	Planned — regenerate mesh (<code>fin_pitch = 4 mm</code>)—	
6 mm pitch (12 fins)	Planned — regenerate mesh (<code>fin_pitch = 6 mm</code>)—	

Next step: parameterize `gen_cases_v2.py`'s `fin_pitch` and `n_fins` per case folder, rerun the 4 mm and 6 mm cases, and compare converged CPU T_{max} across all three pitches to identify the thermal optimum.

6. Limitations

- The fin-pitch sweep is not yet complete — only the 5 mm baseline has been run to convergence; the 4 mm and 6 mm cases are mesh-generation-ready but not yet executed.
- Region coupling uses a simplified coupledTemperature boundary condition rather than a baffle-resistance formulation, so thermal interface material (TIM) contact resistance between the CPU and heatsink is not explicitly represented.
- Outer duct boundaries (`base_bottom`, `top_wall`, `side_walls`) use `noSlip` walls rather than a true open-atmosphere or symmetry condition, which constrains the flow to the duct cross-section.

- The fan curve table is a quadratic fit to the published Delta FFB0812VH curve rather than measured operating data for this exact duct impedance.

7. Methodology Summary

- **Geometry:** Parametric blockMeshDict generation via gen_cases_v2.py (Python) — fin pitch and fin count as direct inputs.
- **Mesh:** blockMesh hex mesh with fin/channel/side-gap strip decomposition; splitMeshRegions -cellZones for fluid/heatsink/cpu region split (649,200 cells total).
- **Region Coupling:** coupledTemperature boundary condition at all fluid↔solid and solid↔solid interfaces.
- **Heat Source:** heatSource fvModel, cellZone allCpuCells, Q = 150 W.
- **Fan Modelling:** fanPressure boundary condition with tabulated quadratic Delta FFB0812VH curve (pressureVsQ.csv).
- **Solver:** foamMultiRun (buoyant fluid + conductive solid, k-ε turbulence), 20,000 steady-state iterations.
- **Verification:** Baseline case run serial and on 8 MPI ranks; converged T_max agrees to 0.001 K.
- **Post-processing:** cpuTmax / solidTmax volFieldValue and surfaceFieldValue function objects; fan patch flow rate and velocity monitors.

Software Stack

Tool	Version	Purpose
OpenFOAM	13	CFD solver (foamMultiRun, buoyant multi-region CHT)
Python	3.x	Parametric blockMeshDict generation
Ubuntu (WSL2)	24.04	Operating system
OpenMPI	—	8-core parallel verification run

8. Conclusion

A steady-state conjugate heat transfer simulation of a CPU and fin-array heatsink has been completed using OpenFOAM 13's foamMultiRun multi-region driver, with a fully parametric mesh generation pipeline built specifically to support a fin-pitch sweep. The 5 mm/15-fin baseline case converges to a CPU temperature of 343.90 K (70.9°C) under 150 W dissipation and fan-curve-driven airflow, with the result cross-checked and confirmed identical between a serial run and an 8-core parallel run.

This work establishes the verified baseline and the parametric infrastructure needed to complete the planned 4 mm / 5 mm / 6 mm fin-pitch comparison and identify the thermal optimum for this fan/duct combination. The full case files, including the parametric mesh generator and solver configuration, are available at github.com/samuelsimmons99/OpenFOAM-Portfolio.